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Problem Statement : Develop a basic software Requirement Specification (SRS) and user stories for a specified software application.

Solution :

A SRS is a formal document that clearly describes what a software system do and how it should behave. It defines all functional and non-functional requirements, constraints, and assumptions of the software system.

In simple words, SRS acts as a blueprint for developers, testers, and stakeholders to understand the software before it is built.

Personal Expense Tracker (PET)

Introduction

Purpose

The purpose of this document is to describe the functional and non-functional requirements of the personal expense tracker system. This SRS serves as a reference for developers, testers, and users.

Scope

The personal expense tracker is a console-based application designed to help users manage their daily expenses. The system allows:

Adding expenses

Viewing expense list

Tracking total spending

Setting monthly budget

Checking remaining balance

Saving and loading data from files.

The system reduces manual calculations and helps users monitor their finances effectively.

Definitions, Acronyms, Abbreviations

Term	Description
PET	Personal Expense Tracker
Budget	Monthly spending limit set by user
Expense	Record containing category, amount, date, description.

Overall Description

Product Perspective

The personal expense tracker is a standalone console application developed in C++. It stores expense data in text files and runs locally on the user's computer.

User Classes and Characteristics.

User	Description
User	Any individual managing personal expenses.

(The system has only one role.)

Operating Environment

OS : windows / linux / mac

Language : C++

Compiler : g++ / CodeBlocks / Visual Studio

Storage : local text files.

Interface : console based.

Constraints

Application runs only in terminal / console

File storage must be accessible

User must enter valid numeric values.

No online / cloud synchronization

Functional Requirements

FR1 : The system shall allow the user to add a new expense.

FR2 : The system shall store category, amount, date, and description.

FR3 : The system shall display all recorded expenses.

FR4 : The system shall calculate total expenses.

FR5 : The system shall allow the user to set or update a monthly budget.

FR6 : The system shall display remaining

balance after expenses.

FR7: The system shall alert the user if the budget is exceeded.

FR8: The system shall save expense data to a file.

FR9: The system shall load saved data when the program starts.

Non-Functional Requirements.

NFR1 (Performance): The system shall respond instantly to user input.

NFR2 (Usability): The system shall provide a simple menu-driven interface.

NFR3 (Reliability): Data shall be stored safely in files.

NFR4 (Maintainability): The code shall follow OOP concepts (encapsulation, inheritance, polymorphism, abstraction).

NFR5 (Portability): The system shall run on any system supporting a C++ compiler.

Assumptions and Dependencies

User has basic knowledge of using console programs.

File system is available for storing data.

C++ compiler is available.

User stories for Personal Expense Tracker.

- Standard Format:

As a <user>, I want <feature>, so that <benefit>.

User Stories:

US1:
As a user, I want to add expenses so that I can record my spending.

US2:
As a user, I want to view all expenses so that I can track where my money goes.

US3:
As a user, I want to see my total spending so that I can understand my monthly expenses.

US4:
As a user, I want to set a budget so that I can control my spending.

US5:
As a user, I want to see my remaining balance so that I know how much I can still spend.

US6:
As a user, I want the system to alert me when I exceed my budget so that I avoid overspending.

US7:
As a user, I want my data saved automatically so that I don't lose my records.

~~Sample user story with acceptance criteria~~
User Story

As a user, I want to add an expense so that I can track my daily spending.

Acceptance Criteria:

Expense is saved when valid details are entered

Amount must be numeric

Expense appears in the expense list

Total updates automatically

Conclusion

This document presents the SRS and user stories for the personal expense tracker system. The SRS formally defines system requirements, while user stories describe them from the user's perspective, supporting both structured and agile development approaches.